MOHAMMAD AMIN TALAGHAT, Ph.D.

AI-Driven Digital Twin & Smart Infrastructure | Pavement Management Systems | BIM/IoT/AI | Construction Management & Contracts

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PROFESSIONAL PROFILE

Civil engineer and construction management professional with a completed Ph.D. in Construction Management, focused on AI-driven Digital Twin systems for pavement and smart infrastructure. Experienced in connecting academic research with product-oriented and commercial infrastructure applications, including road pavement digital twins, automated condition assessment, data-driven maintenance decision support, BIM/IoT integration, and infrastructure asset management. Combines international research collaboration with practical experience in construction project management, technical office leadership, contract administration, delay claims, cost estimation, and web/software development.

TARGET COLLABORATION FIT

- Direct strongest area: highway, road and urban pavement digital twin solutions, including AI-based pavement distress detection/segmentation, condition assessment, PCI-oriented workflows, and maintenance optimization.
- Transferable adjacent area: airport pavement monitoring and digital twin implementation, using the same core logic of georeferenced inspection, computer vision, data fusion, asset management, and decision-support workflows.
- Commercial bridge: ability to support regional infrastructure collaboration by combining research credibility, local project/contract experience, and product-oriented technical communication.

CORE EXPERTISE

Digital Twin for Road Pavement and Infrastructure Assets	AI / Machine Learning / Deep Learning for Condition Assessment
Pavement Management Systems and Maintenance Decision Support	Computer Vision-Based Distress Detection and Segmentation
BIM, IoT Integration and Data-Centric Engineering Workflows	Project Planning, Project Controls and Technical Office Management
Contract Administration, Claims, Delay Analysis and Cost Estimation	Python, Data Mining, Web Development and Engineering Software Tools

EDUCATION

Ph.D. in Civil Engineering - Construction Engineering and Management — **2021 - Jan 2026**
Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran
GPA: 18.98/20 | Thesis: Development of a digital twin-based intelligent decision support system for asphalt pavement maintenance | Supervisor: Dr. Amir Golroo

Ph.D. Visiting Scholar, DigiPave Project — **Oct 2024 - Apr 2025**
University of Oulu, Oulu, Finland
Host Professor: Professor Rauno Heikkila
Research focus: road pavement digital twin, AI-driven infrastructure monitoring and intelligent maintenance systems

M.Sc. in Civil Engineering - Construction Engineering and Management — **2017 - 2020**
K. N. Toosi University of Technology, Tehran, Iran
GPA: 17.91/20 | Thesis: Development of a building energy management system by integrating BIM and IoT to improve efficiency in public buildings | Supervisor: Dr. Ehsan Saghatforoush

B.Sc. in Civil Engineering — **2011 - 2016**
Shiraz University, Shiraz, Iran

PROFESSIONAL AND RESEARCH EXPERIENCE

Technical & Contracts Affairs Manager | Sahar Sazeh Novin Paydar Co. — **Jun 2025 - Present**
Shiraz, Iran

- Manage technical office and contractual documentation, including project correspondence, progress reporting, estimation, contract records, and claim-related evidence.
- Prepare and review project control documents, technical reports, quantity take-offs, interim payment statements, variation/price analysis and contractual deliverables.

Research Collaborator / Research Assistant | University of Oulu - DigiPave Project **Jan 2024 - Present**
Oulu, Finland / Hybrid

- Worked on AI-driven road pavement digital twin architecture, automated condition assessment, data-driven maintenance decision support, and smart infrastructure systems.
- Integrated pavement engineering, georeferenced inspection data, computer vision, AI analytics, BIM/IoT concepts, and maintenance planning into practical digital twin workflows.

Co-Founder, CEO and Project Manager | ArkaNoyan Industry Development Co. **2019 - Present**
Shiraz, Iran

- Led construction-contracting and engineering activities covering project coordination, execution management, technical office work, project planning, cost control and commercial follow-up.
- Coordinated multidisciplinary teams and managed executive, technical, financial and contractual aspects of building and infrastructure-related projects.

Technical Office and Project Controls Supervisor | Arshsazan Negin Co. **Apr 2025 - Jun 2025**
Shiraz, Iran

- Supervised technical office workflows, project controls, documentation, quantity surveying, scheduling, and coordination between site, design and management teams.

Contract Affairs Expert | Iran Gas Engineering and Development Co. / NIGC Eng. **Jan 2022 - Jan 2024**
Tehran, Iran

- Supported contract affairs, risk analysis, contractual documentation, project finance and BOT-related studies for oil, gas and infrastructure project environments.
- Contributed to research and technical studies on fair risk allocation, financing models and contractual structures for infrastructure projects.

Technical & Execution Supervisor | Peynavardan Shahr Co. **2020 - 2021**
Shiraz, Iran

- Supervised execution activities, technical documentation, site coordination and quality-related follow-up for construction works.

Technical Office Engineer and Project Controls Engineer | Debleh Construction Co. **2018 - 2019**
Tehran, Iran

- Prepared schedules, progress control documents, quantity surveying reports, technical submittals and project coordination documents.

Web Developer and Designer / Director | WebAlfa and Tarnamagostar Parslabkhand Co. **2016 - Present**
Shiraz, Iran

- Designed and developed web platforms and applications using UI/UX, HTML/CSS, JavaScript/React, PHP/Laravel and related digital tools.

PROJECTS

- Road Pavement Digital Twin / DigiPave - decision-support system for pavement maintenance, Oulu, Finland.
- Building smartness and BIM/IoT-based energy management system for Roger Williams College, Florida, USA.
- Deep vision-based railway condition assessment using AI and georeferenced maintenance intelligence on the line, Tehran, Iran.
- Delay and damages claim preparation and project scheduling for the 3,300-ton/day Fars Cement Factory in Kharameh, Iran.
- Delay claim documentation for Q Residential-Commercial Complex, 170,000 m2, Khorramabad, Iran.
- Project planning and controls for the 1,210-unit Mehr Housing Project, 120,000 m2, Pardis, Iran.
- Quantity surveying and estimation for the 60-hectare Mehr Housing landscaping project, Pardis, Iran.
- Construction of the Electrical and Computer Engineering Faculty of Shiraz University, 6,000 m2, Shiraz, Iran.
- Design, drawings and structural calculations for a 6-story residential building, 2,000 m2, Shiraz, Iran.
- Design and development of WebAlfa, BIMGate, Paprika and TwinUp digital platforms.

HONORS, LICENSES AND CERTIFICATES

- Ph.D. Visiting Scholar (Funded Fellowship), University of Oulu, Oulu, Finland, Oct 2024 - April 2025
- Member of Iran Construction Engineering Organization; Grade 3 Licensed Construction Supervisor and Executor.
- Winner of Iran's National Elites Foundation talented students' award, 2021.
- Ranked 20th in the Iranian Nationwide Ph.D. Entrance Exam, 2021.
- Ranked 3rd among M.Sc. students in Construction Engineering and Management, 2020.
- Certificates/courses: HSE, MATLAB engineering programming, ETABS/SAFE structural design, AutoCAD, Revit/BIM, Python data mining and machine learning, Microsoft Project, Taksa quantity surveying and payment statements.

TECHNICAL TOOLS AND LANGUAGES

Programming/Data: Python, MATLAB, JavaScript, React.js, PHP/Laravel, HTML/CSS, RapidMiner, Weka, Calude
Engineering/Construction: Microsoft Project, Primavera P6, Microsoft Office, AutoCAD, Civil 3D, Revit, InfraWorks, ETABS, SAFE, Tekla Structures, Taksa
Management/Domain: BIM, IoT, Digital Twin, Pavement Management Systems, Project Control, Contract Management, Risk Management, PMBOK, QA/QC, HSE
Languages: Persian - Native; English - Professional working proficiency

PUBLICATIONS AND MANUSCRIPTS

1. Talaghat, M. A., Golroo, A., Shahhosseini, V., & Rasti, M. (2026). An implemented AI-driven digital twin architecture for proactive pavement maintenance in Finland. *Scientific Reports*.

2. Talaghat, M. A., Golroo, A., Shahhosseini, V., Rasti, M., Daneshvar, A., & Moghadas Nejad, F. (2026). Hybrid MLP–LSTM deep predictive modeling of pavement distress severity and density and condition indices. *International Journal of Pavement Engineering*, *27*(1). 2624565.

3. Talaghat, M. A., Golroo, A., Rasti, M., & Shahhosseini, V. (2026). Road Pavement Digital Twin: A Blueprint with a Case Study. *Journal of Computing in Civil Engineering*, *40*(3), 05026001.

4. Talaghat, M. A., Golroo, A., Shahhosseini, V., & Rasti, M. (2025). Enhancing pavement distress segmentation model with diffusion-based generative AI: a digital twin perspective. *International Journal of Pavement Engineering*, *26*(1), 2540076.

5. Talaghat, M. A., Golroo, A., Kharbouch, A., Rasti, M., Heikkilä, R., & Jurva, R. (2024). Digital twin technology for road pavement. *Automation in Construction*, *168*, 105826.

6. Talaghat, M. A., Golroo, A., Shahhosseini, V., & Rasti, M. (2026). Fully Automated Pavement Crack Quantification and PCI Estimation via a Digital Twin-Ready Vision-Based Deep Learning Pipeline. *Results in Engineering*.

7. Talaghat, M. A., Golroo, A., Shahhosseini, V., Rasti, M., & Daneshvar, A. (2026). A data-driven optimization framework for network-level pavement maintenance planning. *International Journal of Pavement Engineering*. *(under review)*

8. Sedighian-Fard, M., Golroo, A., Javanmardi, M., Alahi, A., Dehghan Tezerjani, H., Talaghat, M. A., Shahhosseini, V., & Rasti, M. (2026). Explainable AI-enhanced pavement deterioration modeling using deep learning. *Scientific Reports*.

9. Nouri, S., Rezaeemanesh, M., Talaghat, M. A., Rasti, M., & Golroo, A. (2026). Self-supervised crack segmentation with data-driven severity detection. *IEEE Transactions on Intelligent Transportation Systems*. *(under review)*

10. Sedighian-Fard, M., Talaghat, M. A., et al. (2026). Deep vision-based railway track superstructure condition assessment using YOLOv26 and georeferenced maintenance intelligence: Field validation on the Zanjan-Tehran railway line. *Scientific Reports*. *(under review)*

11. Sedighian-Fard, M., Golroo, A., Javanmardi, M., Alahi, A., Talaghat, M. A., Shahhoseini, V., Amiri, S., Moghadas Nejad, F. (2026). A complete pipeline for automated pavement distress detection, segmentation, and quantification using a new surface condition index. *Case Studies in Construction Materials. (under review)*

12. Sedighian-Fard, M., Javanmardi, M., Alahi, A., Talaghat, M. A., & Shahhosseini, V. (2026). Vehicle-based pavement condition monitoring: A review of evidence, technology readiness, and management integration. *Journal of Traffic and Transportation Engineering. (under review)*

13. Alobaidi, M., Golroo, A., Sedighian-Fard, M., & Talaghat, M. A. (2026). Synchronized 2D–3D edge sensing for automated ASTM D6433 pavement condition assessment: A Baghdad case study. *Measurement*. *(under review)*

14. Talaghat, M. A., Sedighian-Fard, M., Golroo, A., & Rasti, M. (2025). Leveraging digital twin technology for data-driven pavement maintenance. In *AM3P Conference* (Vol. 4, pp. 604-608). TU Wien, E230-03 Road Engineering.

15. Talaghat, M. A., Golroo, A., & Rasti, M. (2025). Improving Pavement Distress segmentation with Diffusion-Based Generative AI. *In AM3P Conference* (Vol. 4, pp. 579-582). TU Wien, E230-03 Road Engineering.

16. Sedighian-Fard, M., Fathali, M., Yazdani Tabai, S. H., Homavandi, S., Akbari, M., & Talaghat, M. A. (2026). Interpretable machine-vision indices for monitoring sand fouling, ballast whitening and vegetation in climatically diverse Iranian railways. International Journal of Railway Research *(under review).*

17. Saghatforoush, E., Abbasianjahromi, H., Talaghat, M. A., & Kahvandi, Z. (2022). Development of a Management System to Improve the Energy Efficiency of Public Buildings by Integrating IoT and BIM. *Lean Construction Journal*.

18. Talaghat, M. A., & Sedighian-Fard, M. (2026). *A self-calibrating probabilistic digital twin of road pavement networks with decision-centric tasking of autonomous vehicle fleets. (under review)*

19. Talaghat, M. A., & Sedighian-Fard, M. (2026). *A self-updating, two-way building digital twin for grid-interactive operation: Forecasting equipment faults and jointly optimising energy, peak demand, carbon and maintenance*. Sustainable Energy, Grids and Networks. *(under review)*

RESEARCH BACKGROUND AND INTERESTS

- Ph.D. and DigiPave background: digital twin-based intelligent decision-support systems for asphalt pavement maintenance, integrating pavement management systems, georeferenced inspection data, AI-driven condition assessment, automated monitoring and maintenance optimization.
- AI/computer vision interests: pavement distress detection and segmentation, crack quantification, severity/density estimation, PCI-oriented pipelines, diffusion-based generative AI, self-supervised learning, YOLO-based infrastructure inspection, MLP-LSTM prediction and explainable deterioration modelling.
- Asset-management and decision-support interests: data-driven prioritization, network-level maintenance planning, predictive condition modelling, treatment optimization and integration of technical, economic and operational constraints for infrastructure owners.
- BIM/IoT and building-energy research: BIM/IoT-based energy management systems, data-centric engineering workflows, operational performance monitoring and occupant-comfort analysis using meta-synthesis and AHP methods.
- Construction management and contracts: project controls, technical office management, quantity surveying, cost estimation, delay claims, contract administration, project finance, BOT studies and fair contractual risk allocation; current interest is practical deployment of AI-driven Digital Twin and smart asset-management platforms in real infrastructure projects.

REFERENCES

Dr. Amir Golroo - Associate Professor of Civil and Environmental Engineering, Amirkabir University of Technology (Tehran Polytechnic) | agolroo@aut.ac.ir

Dr. Mehdi Rasti - Associate Professor of Information Technology and Electrical Engineering, University of Oulu | mehdi.rasti@oulu.fi

Dr. Rauno Heikkilä - Professor of Digitalized Construction and Mining Operations, Faculty of Technology, University of Oulu; Head of the Structures and Construction Technology Research Unit | rauno.heikkila@oulu.fi

Dr. Vahid Shah Hosseini - Associate Professor of Civil Engineering, Amirkabir University of Technology (Tehran Polytechnic) | shahhosseini@aut.ac.ir

Dr. Ehsan Saghatforoush - Associate Professor of Construction Economics and Management, Wits University | ehsan.saghatforoush@wits.ac.za

Dr. Hamidreza Abbasianjahromi - Assistant Professor of Civil Engineering, K. N. Toosi University of Technology | habasian@kntu.ac.ir

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